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MILITARY SPECIFICATION

HEIGHT FINDER, OBLIQUE PHOTOGRAPH, PHOTOGRAMMETRIC: 50° TO 70° RANGE OF TILT, 9- BY 9-INCH PHOTOGRAPHS, WITH CASE

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers an instrument complete with case for use in the determination of ground elevations from high oblique sterescopic aerial photographs by parallax measurement principles.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

C-F-206 — Felt, Sheet, Wool, Pressed.

GGG-C-105 — Caliper, Micrometer and Gage, Depth, Micrometer.

PPP-F-320 — Fiberboard; Sheet, Stock and Cut Shapes.

PPP-B-601 — Boxes, Wood, Cleated-Plywood.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-P-116 — Preservation, Methods of.

MIL-P-514 — Plates; Identification, Transportation-Data and Blank.

MIL-W-530 — Webbing, Textile, Cotton, General Purpose, Natural or in Colors.

MIL-T-2312 — Treatment, Mildew Resistance for Felt, Wool.

MIL-P-5425 — Plastic, Sheet, Acrylic, Heat Resistant.

MIL-A-8625 — Anodic Coatings, for Aluminum and Aluminum Alloys.

FSC 6675

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MIL–C–12044 — Chests, Plywood.

STANDARDS

MILITARY

MIL–STD–129 — Marking for Shipment and Storage.

MIL–STD–130 — Identification Marking of U. S. Military Property.

DRAWINGS

CORPS OF ENGINEERS

D9141–1 through –25 — Photomapping Equipment—Height Fider, Oblique Photograph, Photogrammetric, With Case.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Association, 1616 P Street, N.W., Washington, D.C.)

AMERICAN IRON AND STEEL INSTITUTE

Steel Products Manual — Stainless and Heat Resisting Steels.

(Application for copies should be addressed to the American Iron and Steel Institute, 350 Fifth Avenue, New York 1, N. Y.)

AMERICAN SOCIETY FOR TESTING MATERIALS

ASTM STD B26 — Aluminum-Base Alloy Sand Castings.

ASTM STD B159 — Phosphor Bronze Wire.

ASTM STD B209 — Aluminum-Alloy Sheet and Plate.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 **Fabrication.** The height finder and the case shall be fabricated as shown on Drawings D9141–1 through –25 and as specified herein.

3.2 The drawings forming a part of this specification are engineering design drawings and are not intended for use as shop drawings. Critical dimensions and tolerances are furnished; however, the contractor is responsible for preparing his own shop drawings. Where tolerances are not shown or where those prescribed could cumulatively result in improper fits, the contractor is responsible for providing tolerances in accordance with industry standards for similar equipment and within those prescribed on the engineering design drawings to insure proper fit, assembly, and operation of the height finder. No deviation from dimensions or tolerances shown on the engineering design drawings is permissible without prior approval of the contracting officer.

3.3 **Material.** Material shall be as specified herein and as shown on the applicable drawings.

3.3.1 *Aluminum castings.* Aluminum castings shall conform to ASTM Standard B26, alloy G4A, condition F.

3.3.2 *Aluminum sheet and strip.* Aluminum sheet and strip shall conform to ASTM

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Standard B209, alloy CG 42A, temper T–4.

3.3.3 *Plastic.* Plastic shall conform to Specification MIL–P–5425, finish A.

3.3.4 *Felt.* Felt shall conform to Specification C–F–206, classification 8R5, and shall be treated for mildew resistance in accordance with Specification MIL–T–2312, type III.

3.3.5 *Webbing.* Webbing shall conform to Specification MIL–W–530, type II, class 4.

3.5.6 *Corrosion-resistant steel.* Corrosion-resistant steel shall conform to the AISI Steel Products Manual, type 303.

3.3.7 *Phosphor bronze.* Phosphor bronze shall conform to Standard ASTM STD B159, alloy A.

3.4 **Graduations and markings.** Graduation lines shall be within 0.005 inch of their true relative position, and the distance between any two adjacent lines shall be correct to within 0.005 inch when measured at a temperature of 77° plus or minus 5° F. All scale graduations, numbers, or other markings shall be readily legible and permanently marked.

3.5 **Crosshair positioning.** When the crosshair on the reticles is tested as specified in 4.4.2.1, there shall be no visible space between the corresponding lines.

3.6 **Radial arm alinement.** In the assembled radial arm, the vertical crosshair on the reticle, the scribed radial line on the plastic window, and the centerline of the slot in the lower radial frame shall be in precise alinement. When the radial arm alinement is tested as specified in 4.4.2.2, there shall be no visible space between the vertical crosshair line on the reticle and the constructed line, and the marked positions of the slot in the lower radial frame shall be displaced by not more than 0.01 inch.

3.7 **Magnifying glasses.** The magnifying glasses shall be fabricated as shown on the drawings.

3.7.1 *Lens or lenses.* The lens or lenses shall be made of good quality clear glass, free from striae, bubbles, waves, scratches, or other defects which will impair the optical qualities when viewed with the unaided eye. The surfaces shall be ground and polished.

3.8 **Micrometer.** The micrometer shall conform to the applicable requirements of Specification GGG–C–105, without ratchet, calibrated 0 to 25 mm. The micrometer shall be modified as shown on the drawings, and the thimble and the barrel shall be graduated so that the scale increases from right to left when observed from the operating position.

3.9 **Case.** The case shall conform to the applicable requirements of Specification MIL–C–12044.

3.10 **Identification marking.** The height finder shall be identified in accordance with Standard MIL–STD–130. Identification plates, if used, shall conform to Specification MIL–P–514.

3.11 **Treatment and finishing.** All aluminum parts shall be anodized in accordance with Specification MIL–A–8625, type II, dyed black for the base, nondyed for all other parts.

3.12 **Technical publications to accompany height finder.** Such technical publications as are specified shall be furnished (see 6.2).

3.13 **Workmanship.** Finished surfaces of the height finder shall be free from visible scratches or other surface defects and shall have a smooth, satin finish.

3.13.1 *Machine work.* Tolerances and gages for metal fits shall conform to the limitations specified herein and on the applicable drawings and otherwise to the standards of accepted commercial practice. Finished con-

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tact and bearing surfaces shall be true and exact.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Height finders not conforming to the performance requirements established in section 3 when inspected as specified in section 4 shall be disassembled in the presence of the Government inspector. Each disassembled part shall be inspected in detail for compliance with the specification and drawings in regard to materials, finishes, dimensions, and tolerances. Height finders not complying with such requirements shall be rejected.

4.2 **Classification of inspection.** Inspection shall be classified as follows:

(a) Production inspection of the height finder.

(b) Inspection of preparation for delivery.

4.3 **Production inspection.**

4.3.1 *Examination.* Each height finder shall be examined as specified in 4.4.1. Presence of one or more defects shall be cause for rejection.

4.3.2 *Tests.* Each radial arm assembly shall be assembled with the applicable photo-table pivot assembly and tested as specified in 4.4.2 prior to attaching to the applicable photo-plate. Failure to pass any test shall be considered as not meeting the performance requirements (see 4.1.1).

4.4 **Inspection procedure.**

4.4.1 *Examination.* The height finders shall be examined for the following defects:

101. Nonconformance to drawings.
102. Material not as specified.
103. Parts or components dirty, damaged, or not properly functioning.
104. Graduations or markings not as specified.
105. Assembly incorrect or incomplete.
106. Treatment or finishing not as specified.
107. Technical publications missing.
108. Workmanship not specified.

4.4.2 *Tests.*

4.4.2.1 *Crosshair position test.* To determine the accuracy to which the reticle crosshair is positioned in the center of rotation of the pivot point, construct a straight line on drawing paper or other drafting medium, and mark a reference point at the approximate center of the line. The line shall be either pencil or ink and shall be between 0.004 and 0.005 inch wide. Clamp the pivot assembly with the radial arm in its normal working position to the edge of a table or other flat surface large enough to support the radial arm throughout an arc of 180°. With the radial arm approximately perpendicular to the edge of the table, precisely aline the constructed line under the horizontal line of the reticle crosshair with the reference point centered at the intersection of the crosshair lines. The reference line shall be held in this position by thumbtacks, tape, or other suitable means. Rotate the radial arm 90° to the right and 90° to the left of

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the original position. In each position, the crosshair lines shall be examined with respect to the construction line. If the lines fail to coincide, the instrument shall have failed the test.

4.4.2.2 *Radial arm alinement test.* To determine the accuracy to which the radial slot, the scribed radial line, and the vertical line of the reticle are positioned on a common centerline, draw a straight pencil or ink line at least 42 inches long and between 0.004 and 0.005 inch wide on drawing paper or other drafting medium. Position the radial arm with the slot approximately in the center of the constructed straight line and the scribed radial line in coincidence with the constructed line. The vertical crosshair line on the reticle must coincide with the constructed line or the instrument shall have failed the test. Trace the outline of the radial slot on the drawing paper with a sharp pencil. Rotate the arm 180° about the approximate center of the slot, and again check the vertical crosshair line for conformance to 3.6. Again trace the outline of the radial slot on the drawing paper with a sharp pencil. Remove the radial arm from the constructed line. Measure the displacement of the traced positions of the slot. Displacements in excess of 0.01 inch shall constitute failure of this test.

4.5 **Inspection of preparation for delivery.** The preservation, packaging, packing, and marking shall be examined and tested to determine compliance with section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 **Preservation and packaging.** Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 *Level A.* Each complete height finder and technical publications contained in the carrying case shall be preserved in accordance with Specification MIL–P–116, method 1A–14. The carrying case shall be the inner container. The case shall be closed, and the carrying strap shall be tightly secured around the case. The outer container shall be a box conforming to Specification PPP–B–621, class 2, style optional, or Specification PPP–B–601, overseas type, style optional. The outer container shall be lined with fiberboard sheets conforming to Specification PPP–F–320, type CF, domestic class, single wall, to protect the barrier from abrasion. The box shall be closed and strapped in accordance with the appendix to the box specification. Strapping shall be zinc coated.

5.1.2 *Level C.* Each complete height finder and technical publications contained in the case shall be preserved and packaged to afford protection against deterioration and damage.

5.2 **Packing.** Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 *Levels A and B.* Each complete height finder preserved and packaged as specified in 5.1.1 shall not require packing.

5.2.2 *Level C.* Each complete height finder preserved and packaged as specified in 5.1.2 shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with Uniform Freight Classification Rules, National Motor Freight Classification Rules, or other carrier rules applicable to the mode of transportation.

5.3 **Marking.** Packages and shipping containers shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 **Intended use.** The height finder is intended for use in the determination of ground elevations from high oblique aerial photographs by the parallax measurement principle.

6.2 **Ordering data.** Procurement documents should specify the following:

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(a) Title, number, and date of this specification.

(b) *Technical publications required* (see 3.12).

(c) Administrative provisions for inspection records (see 4.1).

(d) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

6.3 Invitations and contracts should contain the following information concerning the performance tests of the height finder (see 4.1.1):

"To assure that the end item has been made in strict compliance with the specification and drawings, the end item shall be inspected by the contractor in accordance with the specification. An end item not complying with the performance inspection requirements shall be disassembled by the contractor. Each disassembled part shall be inspected by the contractor in the presence of the Government inspector to determine compliance with the specification and drawings.

"When it is established that the disassembled parts have been made in compliance with the specification and drawings and have been assembled correctly, the Government will either accept the end item or make such design modifications as are necessary to assure that the end item complies with the performance requirements. The Government will perform an additional production inspection on the modified end item. Costs incident to such modifications and production inspections of the modified end item will accrue to the account of the Government."

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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